

10(a)	<i>any three from:</i> 1 change / manipulate / alter / modify, (organism's) DNA / genetic material / genome ; 2 to, change / modify, (organism's) characteristics / phenotype ; 3 new / different / functional, protein produced ; 4 transfer / add / insert, gene / allele / DNA (into a recipient organism) ;	3 IGNORE genetic screening 1 IGNORE genetic information 1 ALLOW recombinant DNA in organism context 1 ALLOW adjust for 'change' 2 ALLOW traits / features, for 'characteristics' 4 ALLOW gene editing / edit genes 4 ALLOW delete / replace, gene (in GMO)
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Question	Answer	Marks	Guidance
10(b)	<i>any six from:</i> <i>prepare the vector</i> 1 (normal / dominant) allele / gene, for ADA / adenosine deaminase ; 2 inserted into, vector / virus ; 3 virus modified so it cannot replicate ; <i>obtain and modify patient cells</i> 4 get (haemopoietic / lymphoid) stem cells from (patient) bone marrow ; 5 mix / culture, (engineered) vector, with, patient's / stem, cells (in vitro / ex vivo) ; 6 drug to increase stem cells (in bone marrow) before mp 4 OR drug / chemotherapy / radiotherapy, to kill stem cells (to make space) before mp 7 ; <i>return modified cells to patient</i> 7 reintroduce (GM / transgenic) cells to, patient / bone marrow / blood ; 8 AVP ; 9 AVP ;	6	1 ALLOW <i>IL2RG / DCLRE1C</i>, for 'gene for ADA' 2 ALLOW retrovirus / lentivirus 2 IGNORE adenovirus / AAV / plasmid 3 ECF if incorrect virus named e.g. adenovirus / AAV 4 ALLOW get T, lymphocytes / cells, from blood 5 ALLOW idea of inserting, virus / vector, into patient's / stem, cells (in vitro / ex vivo) 5 ALLOW if patient cells stated to be T cells 7 ALLOW patient T cells returned to blood 8, 9 gene, cloned in / taken from, GM bacteria multiply transformed, patient / T / stem, cells T-lymphocytes / T-cells, produce ADA (protein / enzyme)